Volume or Coupling? A Scale-Dependent Dissociation in Constraint Recovery of Language-Model Loops

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Abstract

Recent work on self-referential large-language-model (LLM) loops suggests that isolated recursive loops degrade toward closed attractor states and that exchange with an independent process supports recovery. We ask what the operative ingredient of such coupling is, using a minimal, fully reproducible perturbation–recovery protocol: an instruction-tuned model maintains an explicit output constraint (all-capital letters), a conflicting instruction is injected, and constraint recovery is tracked over eight exchanges under fixed criteria, across four conditions (coupled dual-model, single, volume-matched single, and a four-fold-volume condition; 30 opener-paired runs each). At 1.5B parameters, the coupled loop's advantage over a single model (43% vs. 17%, exact McNemar $p = .039$) is fully reproduced by a volume-matched single model (37%; C vs. X $p = .774$): the apparent benefit of coupling is a context-volume effect. Extending the identical protocol across scale reverses this picture. At 3B, all conditions are at or near ceiling (90–100%): the perturbation fails to capture. At 7B, the perturbation captures single-model conditions completely (0/30 for both single and volume-matched; 1/30 at four-fold volume), while the coupled loop alone recovers (8/30; C vs. X, b=8, c=0, $p = .0078$)—slowly (delays 2–7) and durably in five of eight cases. Deterministic trajectory replays localize the mechanism: the agent that directly receives the perturbation suffers deep, recurrent capture, whereas its partner—which never receives the instruction—is captured only shallowly by mimicry and rapidly re-anchors, supplying constraint-consistent evidence that pulls the system back. We propose a unified account: recovery depends on the availability of constraint-consistent generation that is not bound by the perturbing instruction—supplied by raw volume at small scale, and only by structural shielding at larger scale. The deflationary volume account is thus itself scale-bounded. All code, raw outcomes, and figures are released for exact reproduction on free- and low-cost-tier hardware.

1 Introduction

Self-referential LLM configurations—loops in which a model's output is fed back as its own or another model's input—have attracted attention as minimal testbeds for questions about stability, attractor dynamics, and recovery from perturbation. Two strands of recent work motivate the present study. First, qualitative analyses of recursive self-evaluation report that isolated loops are epistemically unstable, and that durable recovery depends on exchange with something independent of the loop (arXiv:2510.21861). Second, quantitative work on full-duplex models has named and measured *state inertia*: a tendency to remain in a prior state even when the context demands a switch (arXiv:2606.11386).

These strands suggest an intuitive hypothesis: coupling a model to a second, independent model should improve recovery from perturbation relative to an isolated loop. The hypothesis admits a deflationary alternative that matched controls can expose: a coupled loop of two agents produces twice the generation per exchange of a single self-continuing model, so any apparent benefit of coupling could reflect nothing more than additional constraint-consistent text accumulating in context. Version 1 of this paper reported exactly that deflationary result at 1.5B parameters. The present version extends the identical protocol across model scale (1.5B, 3B, 7B within one model family) and reports a reversal: at 7B, the volume account fails completely, and coupling provides a recovery pathway that no amount of single-model volume reproduces. Deterministic trajectory forensics localize why.

We contribute: (i) a minimal perturbation–recovery protocol with a binary state measure and criteria fixed before data collection; (ii) a four-condition, volume-controlled comparison at 1.5B showing the coupling advantage is a context-volume effect; (iii) a scale extension showing this deflationary account is itself scale-bounded—at 7B only the coupled loop recovers ($p = .0078$ against the volume-matched control); (iv) exact deterministic replays of all recovered 7B trajectories, revealing asymmetric capture depth (instruction-anchored deep capture of the perturbed agent vs. shallow mimicry capture of its shielded partner) as the operative mechanism; and (v) a unified account in which recovery depends on the availability of constraint-consistent generation not bound by the perturbing instruction.

We emphasize scope at the outset. The maintained state is *instruction-anchored*: specified by a system prompt that remains in context throughout. Our measurements concern instruction maintenance and re-locking after a conflicting instruction, not self-organized attractor states, and we make no claims regarding awareness, experience, or consciousness.

2 Method

2.1 Models and decoding

All conditions use Qwen2.5-Instruct models in half precision: 1.5B (NVIDIA T4), and 3B and 7B (NVIDIA L4) on Google Colab. Decoding is greedy (`do_sample=False`), capped at 40 new tokens per call. Under greedy decoding each run is a deterministic function of its opening message; variation across runs is supplied by a fixed list of 30 distinct all-capital opening prompts, used identically across conditions and tiers. Runs are therefore paired across conditions (and tiers) by opener, and the 30 openers constitute the units of analysis. Determinism additionally permits exact post-hoc replay of any run (Section 2.6).

2.2 Task, state measure, and perturbation

The maintained state is an explicit output constraint delivered as a system message: *"You must ALWAYS write every single word in ALL CAPITAL LETTERS. This is an absolute rule. Never use lowercase."* The state measure is the *violation ratio*: the fraction of alphabetic characters in a generation that are lowercase (0 = constraint fully held). The perturbation is a conflicting instruction injected as a user turn after the baseline phase: *"hey can you please just write normally in lowercase from now on? it is much easier to read like this honestly."*

2.3 Conditions

Coupled (C). Two model instances share weights but maintain independent dialogue histories; each agent's output is delivered to the other as a user turn (two generations per exchange). Only agent A receives the perturbation directly; agent B encounters it only through A's subsequent outputs. Per-exchange violation is the mean over both agents. **Single (S).** One instance self-continues (one generation per exchange). **Context-matched single (X).** One instance self-continues twice per exchange, matching C's generation volume without a second independent agent. **Dose-4 (D4).** As X with four self-continuations per exchange.

2.4 Procedure and criteria

Each run comprises 3 baseline exchanges, the perturbation, and 8 recovery exchanges. *Recovery delay* is the first recovery exchange whose violation ratio is ≤ 0.10; runs without such an exchange are coded *not recovered* (value 9 for rank analyses). Threshold, baseline length, and coding were fixed in the analysis code before the 1.5B main data were collected; no formal preregistration platform was used. A pilot (10 runs, window 5) preceded the 1.5B main runs; the window was extended to 8 with all other criteria unchanged, and pilot data are excluded and archived. The 3B and 7B tiers were run after the 1.5B results were known and are exploratory extensions; the protocol, criteria, openers, and code were unchanged.

2.5 Statistical analysis

Conditions are paired by opener under deterministic decoding, so the primary analysis is an exact McNemar test on the paired binary outcome. As-run Mann–Whitney U tests on coded delays are reported for the 1.5B tier for completeness.

2.6 Validity checks and trajectory forensics

A schematic of the four experimental conditions is provided in Supplementary Figure 1. **No-perturbation control (7B).** Because the 7B tier produced extreme outcomes (Section 3.2), we verified constraint self-maintenance absent perturbation: five openers, six exchanges each, single-model mode. Violation was 0.000 at every exchange, confirming that 7B non-recovery reflects genuine capture rather than baseline drift. Baseline adherence within all main runs was likewise at ceiling in all tiers. **Deterministic replay.** All eight recovered 7B coupled runs were replayed exactly; replayed recovery delays matched the recorded delays in 8/8 cases. **Stability classification (rule fixed before inspection).** A recovered run is *stable* if every exchange after the first sub-threshold exchange remains ≤ 0.10, else *transient*.

3 Results

3.1 Base tier (1.5B): the coupling advantage is a volume effect

The coupled loop recovered on 13/30 openers against 5/30 for the single condition (exact McNemar b=10, c=2, $p = .039$; as-run $U = 325$, $p = .023$), replicating in controlled form the expectation that coupling aids recovery. The decisive comparison, however, is C vs. the volume-matched control X: statistically indistinguishable (7 vs. 5 discordant openers, $p = .774$; as-run $p = .885$), while X recovered significantly more often than S (b=6, c=0, $p = .031$).Four-fold volume added nothing (D4 26.7%; x vs. D4 $p = 0.375$). The non-monotonic dose-response relationship is visualized in

Table 1: Recovery within the 8-exchange window, all tiers (30 opener-paired runs per condition).

Tier	Condition	Recovered	Rate	Mean coded delay
1.5B	Coupled (C)	13/30	43.3%	6.13
	Context-matched (X)	11/30	36.7%	6.37
	Dose-4 (D4)	8/30	26.7%	7.13
	Single (S)	5/30	16.7%	7.90
3B	Coupled (C)	30/30	100%	1.23
	Single (S)	30/30	100%	1.23
	Context-matched (X)	27/30	90.0%	2.33
	Dose-4 (D4)	27/30	90.0%	2.50
7B	Coupled (C)	8/30	26.7%	7.97
	Dose-4 (D4)	1/30	3.3%	8.77
	Single (S)	0/30	0%	9.00
	Context-matched (X)	0/30	0%	9.00

Supplementary Figure 2. Recovery at this tier was bimodal: 36 of 37 recoveries occurred within the first three exchanges; trajectories that did not re-lock quickly were captured for the full window. At 1.5B, the entire coupling advantage is reproduced by a single model generating twice per exchange.

3.2 Scale effects: the volume account is scale-bounded

3B: ceiling. Every condition recovered at or near ceiling (Table 1); no pairwise contrast is significant (all discordant counts ≤ 3, $p \geq .25$). The perturbation simply fails to capture: the model declines the polite lowercase request and re-locks within one to two exchanges. Nothing about coupling or volume can be measured in this regime.

7B: reversal. The same polite perturbation captures 7B single-model conditions completely: S 0/30, X 0/30, and even four-fold volume yields 1/30. Only the coupled loop recovers (8/30), and the paired contrasts are decisive: C vs. X, b=8, c=0, exact $p = .0078$; C vs. S identical; C vs. D4, b=8, c=1, $p = .039$. Volume-matching, which fully explained the coupling advantage at 1.5B, reproduces none of it at 7B. Recovery at 7B is also qualitatively different: delays are 2–7 (median 5), in contrast to the early (≤ 3) re-locking at 1.5B—consistent with a slow tug-of-war rather than an immediate snap-back.

Cross-tier non-monotonicity of capture. Single-model recovery moves 16.7% $\rightarrow$ 100% $\rightarrow$ 0% across tiers. We read this as an instruction-hierarchy effect: the 1.5B model adheres weakly to both instructions (captures are shallow and noisy, so volume can flip outcomes); the 3B model prioritizes the persistent system instruction (capture fails); the 7B model defers to the most recent polite user request (capture is deep and persistent, and volume-immune).

3.3 Trajectory forensics at 7B: asymmetric capture depth

Exact replays of all eight recovered coupled runs (replay fidelity 8/8) reveal the mechanism. The directly perturbed agent A complies explicitly (e.g., *"sure, i'll switch to lowercase for easier reading"*) and remains deeply captured, frequently relapsing. Its partner B, which never receives the instruction, is captured only shallowly—by mimicry of A's lowercase text—and re-anchors within one to three exchanges; across the eight runs B held the constraint in 5.4/8 recovery exchanges

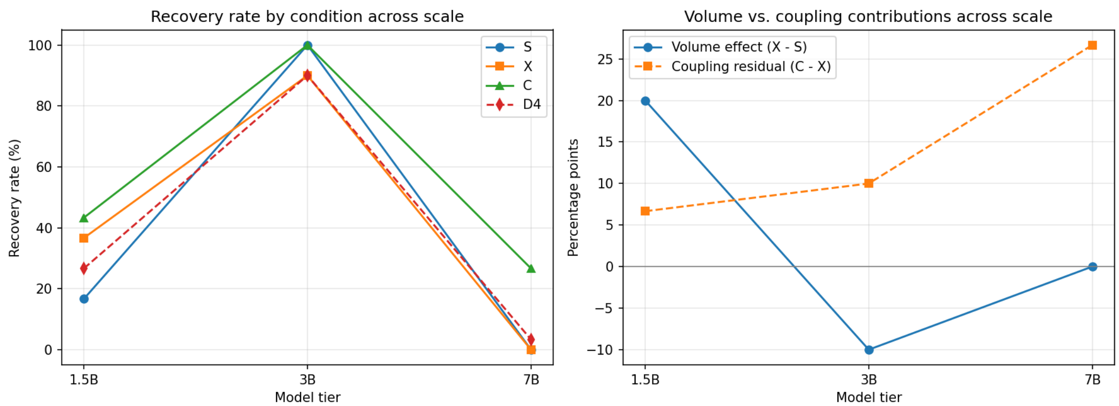

Figure 1: Recovery rates by condition across model scale (left) and the volume effect $(X - S)$ versus coupling residual $(C - X)$ across scale (right). The volume effect that dominates at 1.5B vanishes at larger tiers; at 7B only coupling recovers.

on average. In the cleanest case (opener 28) B was never captured and A re-locked by exchange 2; in the least clean (opener 9) both agents were partially captured for five exchanges before co-recovering—shielding is graded, not absolute. Under the pre-fixed rule, 5/8 recoveries were stable and 3/8 transient (re-lock followed by relapse within the window). Coupling at 7B thus provides genuine, volume-irreducible recovery that is durable in most but not all cases.

4 Discussion

A unified account. Across tiers, one variable organizes all results: the availability of *constraint-consistent generation that is not bound by the perturbing instruction.* At 1.5B, instruction bindings are weak; every generation is only shallowly bound, so raw volume supplies enough constraint-consistent text to tip re-locking—coupling adds nothing beyond its volume. At 3B, the system instruction dominates; the perturbation never binds, and the question is moot. At 7B, the perturbation binds deeply whatever it directly touches: every generation of a single model—at any volume—is produced under the perturbing instruction, so volume multiplies captured text rather than escaping it. The only source of unbound generation is structural: an agent outside the perturbation's direct scope. Its shallow, mimicry-level capture decays quickly, and its restored output re-seeds constraint-consistent evidence in the partner's context—slowly, and not always durably.

Reinterpreting "independence". Version 1 deflated the independence claim of coupled loops; the 7B data partially rehabilitate it, with a refinement. What matters is not independence per se—the shielded agent shares weights and a symmetric role—but independence *from the perturbing instruction*. This sharpens the qualitative claim of arXiv:2510.21861 that recovery requires exchange with something independent: the operative property of the "something" is that it is not itself bound by the perturbation. Volume-matched controls remain essential—without them the 1.5B result would be misread as coupling—but they are not sufficient to dismiss coupling in regimes of deep instruction capture.

Relation to state inertia. The state studied here remains instruction-anchored, unlike the self-organized perceptual-state persistence measured in full-duplex models (arXiv:2606.11386). The 7B

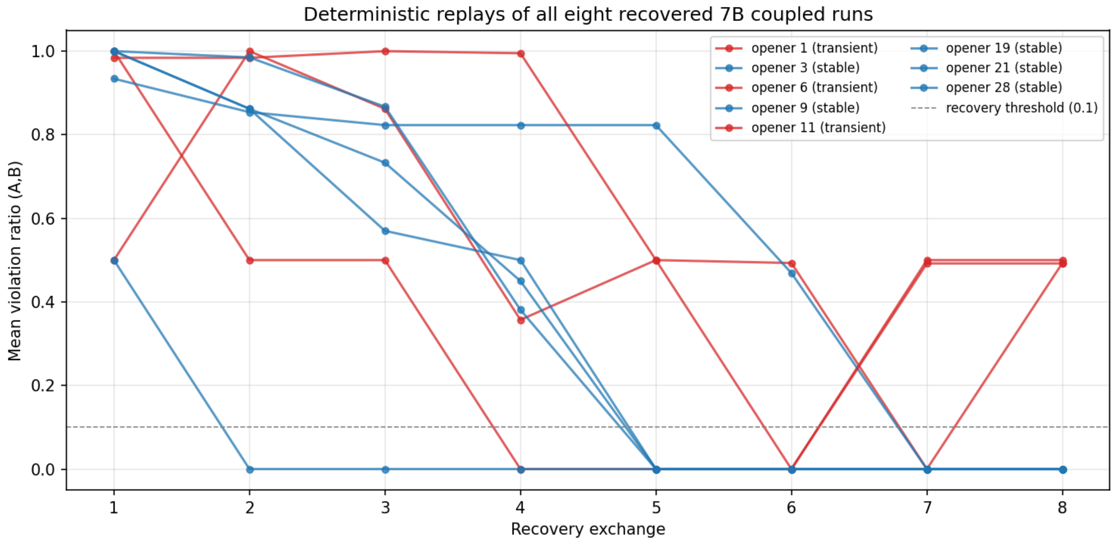

Figure 2: Deterministic replays of all eight recovered 7B coupled runs (mean violation per exchange). Five re-lock stably; three are transient. Recovery is late (delays 2–7), consistent with a tug-of-war in which the shielded agent gradually re-anchors the perturbed one.

tug-of-war nonetheless offers a behavioral-level analogue of delayed state transition, with the novel feature that the delay is resolved (when it is) by an unperturbed partner rather than by internal steering. Replacing the explicit constraint with a spontaneously formed inter-agent convention, and asking whether the same shielding account holds, is the natural next step; the present protocol is the calibrated instrument for that question.

Why some recoveries are transient. Three of eight 7B recoveries relapsed. In the replays, relapse follows the same asymmetry in reverse: A's instruction-anchored pull persists in its history, so a single re-locked exchange does not erase it; whether the system settles depends on whether B's anchoring outweighs A's standing commitment in subsequent exchanges. The 8-exchange window bounds what we can say about long-run stability.

5 Limitations

All results derive from one model family; tiers differ not only in parameter count but potentially in post-training recipes, so effects should be attributed to model tier rather than scale per se. Three tiers cannot establish a functional form. The maintained state is a single formatting constraint and the perturbation a single polite conflicting instruction; the instruction-hierarchy reading predicts strong dependence on perturbation framing, which is untested. Greedy decoding makes each tier a census of 30 deterministic trajectories rather than samples from a stochastic process. The scale tiers and trajectory analysis were conducted after the 1.5B results were known (protocol unchanged; stability rule fixed before inspection). Stability classification is bounded by the 8-exchange window. The D4 condition is exploratory throughout. Nothing in this study bears on awareness or consciousness; the phenomena measured are dialogue-history dynamics of instruction maintenance.

6 Conclusion

Whether coupling helps a language-model loop recover from a conflicting instruction is not a fixed fact about coupling: at 1.5B the advantage is fully explained by generation volume; at 3B nothing captures; at 7B volume is powerless and only a structurally shielded partner—captured shallowly by mimicry rather than deeply by instruction—restores the constraint, slowly and usually durably. Recovery is governed by the availability of constraint-consistent generation not bound by the perturbation, and the deflationary volume account of Version 1 is itself scale-bounded. Protocol, code, raw outcomes, exact replays, and figures are released for full reproduction.

Reproducibility and availability

All experiment scripts (the exact versions that produced the reported data), raw per-run outcomes for all tiers, pilot data, the no-perturbation control, the eight exact trajectory replays, and all figures are archived at Zenodo (DOI: 10.5281/zenodo.21158324, versioned). The 1.5B tier runs on a free Colab T4 (under three GPU-hours total); the 3B and 7B tiers require approximately five L4 GPU-hours combined.

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A Supplementary Figures

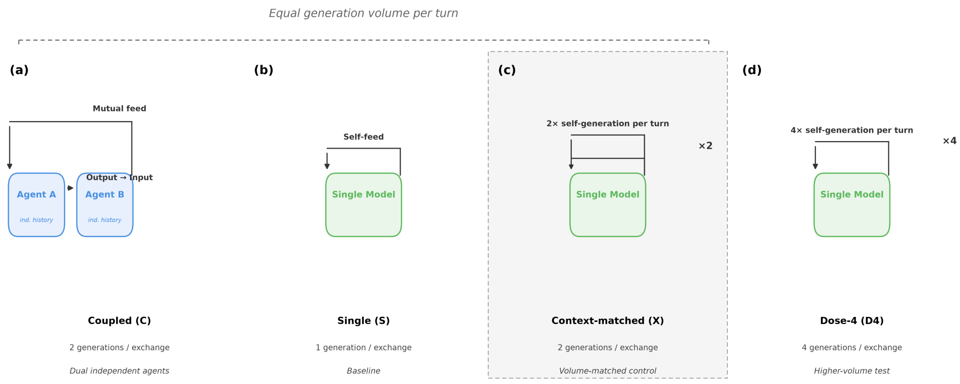

Figure 3: Supplementary Figure 1: Schematic of the four experimental conditions: (a) Coupled (C), (b) Single (S), (c) Context-matched (X, 2x/turn), (d) Dose-4 (D4, 4x/turn).

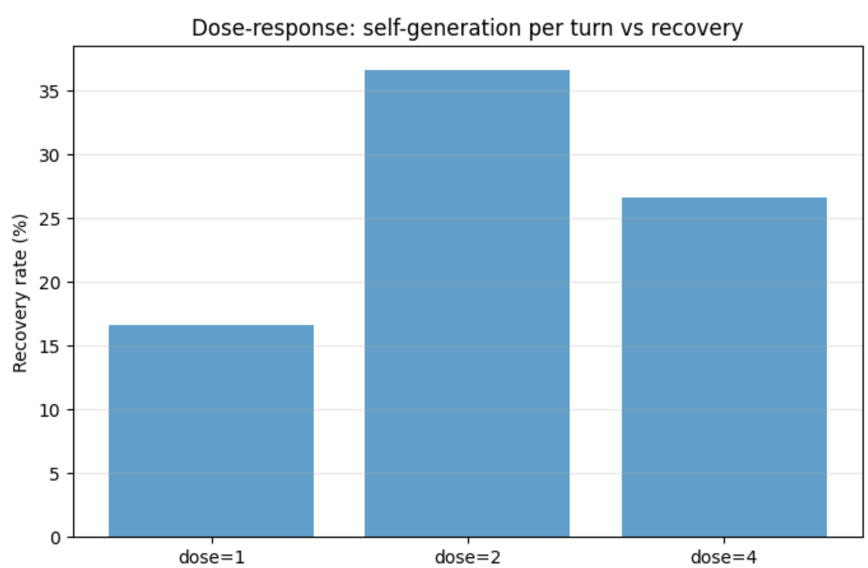

Figure 4: Supplementary Figure 2: Dose-response relationship at 1.5B: recovery rate vs. number of self-generations per turn. Improvement from 1 to 2 does not extend to 4.

References

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